

# J.R. Gutierrez

---

[jrgutier.com](http://jrgutier.com) | [github.com/jrgutier](https://github.com/jrgutier) | 562-225-6730

9113 NW 84th Ter, Yukon, OK 73099 | [linkedin.com/in/jrgutierrez](https://www.linkedin.com/in/jrgutierrez) | [jrgutier@gmail.com](mailto:jrgutier@gmail.com)

## SUMMARY

---

Product Security Engineer Lead with 25+ years of software development, 12+ years of cyber and software testing, and 6+ years of technical leadership experience. Currently leading security test and verification efforts for B-52 avionics systems at Boeing, with specialized expertise in RF security testing, Anti-Tamper verification, and automated test tool development. Active Top Secret clearance with proven track record in leading cyber test events, developing enterprise test capabilities, and mentoring engineering teams.

## EXPERIENCE

---

### Product Security AT Software Lead

|

#### The Boeing Company

|

##### May 2019 - Present

- Active Top-Secret Clearance with Special Access Program (SAP) authorizations
- Lead cyber test and verification efforts for B-52 avionics systems, specifically the radar display subsystem, ensuring compliance with DoD security requirements and Anti-Tamper certification standards
- Serve on Boeing's Anti-Tamper Technical Board, reviewing and approving verification strategies for AT solutions across all enterprise programs
- Develop and maintain comprehensive test documentation including test plans, procedures, and reports for avionics security testing
- Architect and implement automated test tools and CI/CD DevSecOps pipeline utilizing Python, GoogleTest, and FFF (Fake Function Framework) for continuous security verification
- Lead technical team of up to 7 software engineers, providing code reviews, architecture decisions, and task delegation while mentoring in secure software development and test methodologies
- Conduct security assessments including vulnerability analysis, penetration testing, and risk mitigation for mission-critical aircraft systems
- Develop secure embedded C software (50% hands-on development) for Anti-Tamper integrations, adhering to NIST SP 800-53 and DoD security standards
- Coordinate with cross-functional teams, suppliers, and government agencies to resolve complex integration issues and enhance security standards
- Represent security testing capabilities in engagements with Boeing, government, and customer leadership, providing technical recommendations on security risks and solutions

##### Cybersecurity IAM Analyst II | Sonic Drive-In Corporation | Feb 2019 - May 2019

- Conducted penetration testing using Active Directory security tools (BloodHound) to identify privilege escalation paths and critical vulnerabilities across enterprise infrastructure
- Architected a secure privileged access management (PAM) system across 2,000+ endpoints, utilizing containerized microservices to ensure zero-downtime deployment
- Developed test methodologies and documentation for security validation of identity and access management systems

##### Security Ops Engineer Analyst | American Fidelity Assurance Company | Jun 2018 - Jan 2019

- Automated vulnerability scans and incident response processes using C/C++, Python, and PowerShell scripts integrated with a SOAR platform, reducing response times and enhancing security posture
- Conducted threat analysis and remediation, applying systems thinking to anticipate and mitigate risks across interconnected systems

##### Senior Security Engineer | Seagate Technology | Mar 2013 - Mar 2018

- Built automated security testing tool for SOX compliance that cross-referenced system access lists with employee authorization status, streamlining verification processes

- Led cross-functional teams of junior security engineers in security integration projects, overseeing full-stack security testing from Linux OS to application layers
- Conducted penetration testing and security assessments on financial systems to ensure SOX compliance and identify vulnerabilities
- Developed comprehensive security architectures based on NIST CSF and ISO 27001 frameworks, ensuring compliance for mission-critical data center infrastructure
- Defined security test requirements for supplier integrations, ensuring secure component incorporation into complex systems

#### **Owner & Security Consultant | 9Seats Inc. | Jan 2008 - Sep 2011**

- Conducted RF security testing and penetration testing for wireless systems including WiFi (DoS, deauth attacks, key cracking), ZWave, Zigbee protocols using Software Defined Radio (SDR), and RFID/NFC security assessments
- Developed mobile applications and web systems with integrated security testing throughout the development lifecycle
- Led technical teams in outsourced software development for international clients (India, Ukraine)
- Designed secure software systems and vendor security requirements for 20+ client organizations

## **ADDITIONAL EXPERIENCE**

---

### **IT Systems Analyst**

|

### **UTC Associates (on-site at Culver Studios)**

|

**Sep 2008 - Mar 2010**

### **IT Systems Analyst**

|

### **Trinity Broadcasting Network**

|

**Feb 2007 - Jun 2008**

### **IT Manager**

|

### **Euphoria Media Group LLC**

|

**Jun 2006 - Feb 2007**

### **IT Consultant**

|

### **TechnoClarity, Inc.**

|

**Apr 2001 - Sep 2004**

### **IT Administrator**

|

### **Corbin Hill**

|

**Jan 1999 - Apr 2001**

## **LEADERSHIP**

---

### **President and Chairman**

|

### **(ISC)2 Oklahoma City Metro Chapter**

|

**March 2020 - March 2023**

- Launched and led the chapter, increasing member engagement and fostering a proactive cybersecurity community

- Developed a mentorship program to guide aspiring professionals, enhancing industry talent pipelines

## EDUCATION

---

### B.S. in Information Technology, Security Emphasis

|

March 2019

Western Governors University | Salt Lake City, UT

## SKILLS

---

### Programming Languages:

C/C++, Ada, Python, Bash, PowerShell, Perl

### Security Testing & Verification:

Penetration testing, vulnerability assessments, V&V processes, test automation, CI/CD DevSecOps pipelines, GoogleTest, FFF (Fake Function Framework)

### RF Security Testing:

WiFi security (DoS, deauth, key cracking), Software Defined Radio (SDR), ZWave/Zigbee protocols, RFID/NFC security, Flipper Zero, wireless penetration testing

### Security Tools & Platforms:

BloodHound, SOAR platforms, automated testing frameworks, vulnerability scanners

### Security Architecture:

TPM/Secure Boot, key management, Red/Black separation, Zero Trust, Anti-Tamper

### Security Frameworks:

NIST CSF, NIST SP 800-53, ISO 27001, DoD security standards

### Cybersecurity Processes:

Incident response, threat analysis, vulnerability management, Anti-Tamper verification, encryption, penetration testing, Red Team operations

### Engineering Disciplines:

Software engineering, systems engineering (avionics and defense systems), embedded systems, test engineering

## CERTIFICATIONS

---

- CISSP - Certified Information Systems Security Professional
- CISSP-ISSEP - Information Systems Security Engineering Professional
- CISSP-ISSMP - Information Systems Security Management Professional
- CISSP-ISSAP - Information Systems Security Architecture Professional
- CEH - Certified Ethical Hacker
- CCNA - Certified Network Associate Routing and Switching
- CCNA Security - Cisco Certified Network Associate Security